

# Ryoma Otsuka 大塚 亮真



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## Research Interests | 研究の関心

bio-logging, machine learning, deep learning, behaviour recognition/classification, time-series sensor data analysis, mountain gorillas, conservation, wildlife tourism

## Work Experience | 職歴

- |              |                                                                                                                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025-Present | Graduate School of Environmental Studies, Nagoya University<br>Specially Appointed Assistant Professor<br>名古屋大学大学院環境学研究科 特任助教（大阪大学とのクロスアポイントメント）                                                              |
| 2022-Present | Graduate School of Information Science and Technology, Osaka University<br>Specially Appointed Assistant Professor<br>大阪大学大学院情報科学研究科 マルチメディア工学専攻 特任助教                                                         |
| 2021-2022    | Wildlife Research Center, Kyoto University (Postdoctoral Researcher)<br>京都大学野生動物研究センター 研究員（非常勤）<br>Center for African Area Studies, Kyoto University (Postdoctoral Researcher)<br>京都大学アフリカ地域研究資料センター 研究員（非常勤） |

## Education | 学歴

- |                 |                                                                                                                                                                                                                         |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ph.D. 2016-2021 | Graduate School of Asian and African Area Studies (5-year system),<br>Kyoto University, Japan   Ph.D. (Area Studies)  <br>京都大学大学院アジア・アフリカ地域研究研究科（5年一貫制）<br>アフリカ地域研究専攻<br>博士（地域研究）（2021年03月23日）<br>修士（地域研究）（2018年03月26日） |
| B.Sc. 2012-2016 | Faculty of Agriculture, Kyoto University, Japan   B.Sc. (Agriculture)  <br>京都大学農学部資源生物科学科<br>学士（農学）（2016年03月24日）                                                                                                        |

## Research Skills | 研究スキル

- Arduino (or C/C++): to develop custom-made bio-logging devices with AI on-board
- Python: machine learning (scikit-learn), deep learning (PyTorch), Bayesian statistical analysis, time-series data analysis, data visualization
- R: Bayesian statistical analysis (RStan), data visualization
- Fieldwork (Bwindi in Uganda, Aomori, Niigata and Nagasaki in Japan)

## Preprints | プレプリント

Otsuka, R., Kalema-Zikusoka, G., Yamakoshi, G., & Kinoshita, K. (2026). Tourism impacts on fecal glucocorticoid metabolites of free-ranging mountain gorillas in Bwindi Impenetrable National Park, Uganda. *bioRxiv*. <https://doi.org/10.64898/2026.01.15.699166>

## Peer-Reviewed Publications | 査読付き論文

Otsuka, R., Sugiyama, H., Mizutani, Y., Yoda, K., & Maekawa, T. (2025). Real-time behaviour recognition on bio-loggers enables autonomous audio playback experiments in free-ranging seabirds. *Ecology and Evolution*, 15(8), e71832. <https://doi.org/10.1002/ece3.71832>

Otsuka, R., Yoshimura, N., Tanigaki, K., Koyama, S., Mizutani, Y., Yoda, K., & Maekawa, T. (2024). Exploring deep learning techniques for wild animal behaviour classification using animal-borne accelerometers. *Methods in Ecology and Evolution*, 15(4), 716–731. <https://doi.org/10.1111/2041-210X.14294>

Tanigaki, K., Otsuka, R., Li, A., Hatano, Y., Wei, Y., Koyama, S., Yoda, K., & Maekawa, T. (2024). Automatic recording of rare behaviors of wild animals using video bio-loggers with on-board light-weight outlier detector. *PNAS Nexus*, 3(1), pgad447. <https://doi.org/10.1093/pnasnexus/pgad447>

Sekino, A., Otsuka, R., & Yasuoka, H. (2023). Haphazard Sharing of Plant Food among the Baka Hunter-Gatherers in Southeast Cameroon. *African Study Monographs. Supplementary Issue.*, 62, 81–103. <https://doi.org/10.14989/286829>

Maekawa, T., Xia, Q., Otsuka, R., Yoshimura, N., & Tanigaki, K. (2023). Recent Trends in Sensor-based Activity Recognition. *2023 24th IEEE International Conference on Mobile Data Management (MDM)*, 36–38. <https://doi.org/10.1109/MDM58254.2023.00018>

Otsuka, R., Yamakoshi, G., & Kalema-Zikusoka, G. (2023). Tourist expectations and satisfaction in mountain gorilla tourism in Bwindi Impenetrable National Park, Uganda. *Journal of Ecotourism*, 22(2), 329–338. <https://doi.org/10.1080/14724049.2023.2166056>

Hongo, S., Dzefack, Z. C. B., Vernyuy, L. N., Minami, S., Mizuno, K., Otsuka, R., Hiroshima, Y., Djiéto-Lordon, C., Nakashima, Y., & Yasuoka, H. (2022). Predicting bushmeat biomass from species composition captured by camera traps: Implications for locally based wildlife monitoring. *Journal of Applied Ecology*, 59(10), 2567–2580. <https://doi.org/10.1111/1365-2664.14257>

Kano, F., Furuichi, T., Hashimoto, C., Krupenye, C., Leinwand, J. G., Hopper, L. M., Martin, C. F., Otsuka, R., & Tajima, T. (2022). What is unique about the human eye? Comparative image analysis on the external eye morphology of human and nonhuman great apes. *Evolution and Human Behavior*, 43(3), 169–180. <https://doi.org/10.1016/j.evolhumbehav.2021.12.004>

- (in Japanese) 山梨裕美, 徳山奈帆子, 竹ノ下祐二, 大塚亮真, 森村成樹, 赤見理恵. (2021) 大型類人猿と人の関わりの変遷: 過去・現在・そして未来に向けて. *霊長類研究*, 37(2), 155–159. <https://doi.org/10.2354/psj.37.046>
- Costa, R., Tomonaga, M., Otsuka, R., Huffman, M. A., Bercovitch, F., Kalema-Zikusoka, G., & Hayashi, M. (2021). The dispersal dilemma among female mountain gorillas: Risk infanticide and gain protection. *African Journal of Ecology*, 59(1), 273–276. <https://onlinelibrary.wiley.com/doi/10.1111/aje.12824>
- Hanya, G., Tackmann, J., Sawada, A., Lee, W., Pokharel, S. S., de Castro Maciel, V. G., Toge, A., Kuroki, K., Otsuka, R., Mabuchi, R., Liu, J., Hatakeyama, M., Yamasaki, E., von Mering, C., Shimizu-Inatsugi, R., Hayakawa, T., Shimizu, K. K., & Ushida, K. (2020). Fermentation Ability of Gut Microbiota of Wild Japanese Macaques in the Highland and Lowland Yakushima: In Vitro Fermentation Assay and Genetic Analyses. *Microbial Ecology*, 80(2), 459–474. <https://dx.doi.org/10.1007/s00248-020-01515-8>
- Otsuka, R., & Yamakoshi, G. (2020). Analyzing the popularity of YouTube videos that violate mountain gorilla tourism regulations. *PLoS ONE*, 15(5), e0232085. <https://dx.plos.org/10.1371/journal.pone.0232085>

## Grants and Fellowships | 競争的資金の獲得

2018-2021

国家戦略分野の若手研究者及び博士後期課程学生の育成事業 (BOOST)  
次世代 AI 人材育成プログラム (若手研究者支援) (年間 1,000 万円)

2018-2021

Doctoral Course Student (DC1) Research Fellowship, Japan Society of the Promotion of Science (JSPS) (Grant Number: 18J22882) (\$85,454 for 3 years, rate: 1 USD = 110 JPY)  
日本学術振興会特別研究員 DC1 (課題番号: 18J22882) 「マウンテンゴリラの住民参加型保全とエコツーリズムの持続可能性に関する研究」 (940 万円)

2016-2021

Leading Graduate Program in Primatology and Wildlife Science, Kyoto University (\$9,090 annually, rate: 1 USD = 110 JPY)  
京都大学霊長類学・ワイルドライフサイエンス・リーディング大学院  
(特別履修) (年間 約 100 万円)

## Invited Talks | 招待講演

TBA

## Selected Presentations | 学会発表

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## Teaching and Mentoring | 教育と学生指導

### Guest Lectures | ゲスト講義

- 2021-06-07      Area Studies and Tourism in Africa (June 7th) - Hannan University  
                    阪南大学「アフリカの地域と観光」 (06 月 07 日)
- 2021-07-27      Environmental Anthropology (July 27th) - Ryukoku University  
                    龍谷大学「環境人類学」 (07 月 27 日)
- 2021-11-04      Biology B (November 4th) - Chukyo University  
                    中京大学「生物学 B」 (11 月 04 日)
- 2021-11-05      Anthropology of Tourism (November 5th) - Rikkyo University  
                    立教大学「観光人類学 8 (生態)」 (11 月 05 日)

### Mentorship | 学生指導

- 2023-present      Guidance on the development of AI bio-logging devices. Programming instruction for data analysis and implementation of machine learning models. Supervision of undergraduate thesis writing.
- 2022-present      Guidance on the development of AI bio-logging devices. Supervision on data analysis.

## Society Membership

British Ecological Society

## Review Experience | 査読経験

Methods in Ecology and Evolution  
Movement Ecology  
Animal Behaviour  
Primates  
Journal of Ecotourism  
Asian and African Area Studies  
IPSI Transactions on Bioinformatics (TBIO)

## Outreach | アウトリーチ活動

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